

EX No:1	EDA-DATA CLEANING

Aim:

To perform Exploratory Data Analysis (EDA) and data cleaning on the hotel bookings dataset using standard data science frameworks (CRISP-DM, SEMMA, and KDD) by handling missing values, removing duplicates, creating derived features, and analyzing data patterns.

Algorithm:

1. Import pandas, numpy, and matplotlib.pyplot.
2. Load hotel_bookings.csv and inspect dataset structure, missing values, and duplicates.
3. Clean data by removing duplicates, dropping the company column, and imputing missing values in agent, country, and children.
4. Convert reservation_status_date to datetime format and engineer new features (total_nights, total_guests).
5. Plot frequency distributions for key variables and display a correlation matrix heatmap.
6. Summarize business insights and end execution

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("hotel_bookings.csv")

print(df.head())
print(df.shape)
print(df.info())
print(df.describe())
print(df.isnull().sum())
print(df.duplicated().sum())

# 3. DATA PREPARATION

df.drop_duplicates(inplace=True)
df.drop(columns='company', inplace=True)

df['agent'] = df['agent'].fillna(df['agent'].median())
df['country'] = df['country'].fillna(df['country'].mode()[0])
```

```
df['children'] = df['children'].fillna(0)

df['reservation_status_date'] = pd.to_datetime(df['reservation_status_date'])

df['total_nights'] = df['stays_in_weekend_nights'] + df['stays_in_week_nights']
df['total_guests'] = df['adults'] + df['children'] + df['babies']
```

4. MODELING (EDA)

```
df['hotel'].value_counts().plot(kind='bar')
plt.title("Hotel Type")
plt.show()

df['is_canceled'].value_counts().plot(kind='bar')
plt.title("Booking Cancellation")
plt.show()

df['arrival_date_month'].value_counts().plot(kind='bar')
plt.title("Monthly Bookings")
plt.show()

df['country'].value_counts().head(10).plot(kind='bar')
plt.title("Top Booking Countries")
plt.show()
```

```
corr = df.select_dtypes(include='number').corr()

plt.figure(figsize=(10,8))
plt.imshow(corr)
plt.colorbar()
plt.xticks(range(len(corr.columns)),corr.columns,rotation=90)
plt.yticks(range(len(corr.columns)),corr.columns)
plt.title("Correlation Heatmap")
plt.show()
```

5. EVALUATION

```
print("Business insights generated successfully.")
```

6. DEPLOYMENT

```
print("Insights can be deployed through dashboards.")
df = pd.read_csv("hotel_bookings.csv")
```

EXPLORE

```
print(df.head())
print(df.info())
print(df.describe())
print(df.isnull().sum())
df.drop_duplicates(inplace=True)
df.drop(columns='company', inplace=True)
```

```
df['agent'] = df['agent'].fillna(df['agent'].median())
df['country'] = df['country'].fillna(df['country'].mode()[0])
df['children'] = df['children'].fillna(0)

df['reservation_status_date'] = pd.to_datetime(df['reservation_status_date'])

df['total_nights'] = df['stays_in_weekend_nights'] + df['stays_in_week_nights']
df['total_guests'] = df['adults'] + df['children'] + df['babies']

# MODEL

df['hotel'].value_counts().plot(kind='bar')
plt.show()

df['is_canceled'].value_counts().plot(kind='bar')
plt.show()

df['arrival_date_month'].value_counts().plot(kind='bar')
plt.show()

df['country'].value_counts().head(10).plot(kind='bar')
plt.show()

corr = df.select_dtypes(include='number').corr()

plt.figure(figsize=(10,8))
plt.imshow(corr)
plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
plt.yticks(range(len(corr.columns)), corr.columns)
plt.show()

# ASSESS
print("Patterns and trends successfully identified.")

# SELECTION

df = pd.read_csv("hotel_bookings.csv")

# PREPROCESSING

print(df.info())
print(df.isnull().sum())
print(df.duplicated().sum())

df.drop_duplicates(inplace=True)
df.drop(columns='company', inplace=True)

df['agent'] = df['agent'].fillna(df['agent'].median())
df['country'] = df['country'].fillna(df['country'].mode()[0])
df['children'] = df['children'].fillna(0)
```

TRANSFORMATION

```
df['reservation_status_date']=pd.to_datetime(df['reservation_status_date'])
```

```
df['total_nights']=df['stays_in_weekend_nights']+df['stays_in_week_nights']  
df['total_guests']=df['adults']+df['children']+df['babies']
```

DATA MINING

```
df['hotel'].value_counts().plot(kind='bar')  
plt.show()
```

```
df['is_canceled'].value_counts().plot(kind='bar')  
plt.show()
```

```
df['arrival_date_month'].value_counts().plot(kind='bar')  
plt.show()
```

```
df['country'].value_counts().head(10).plot(kind='bar')  
plt.show()
```

```
corr=df.select_dtypes(include='number').corr()
```

```
plt.figure(figsize=(10,8))  
plt.imshow(corr)  
plt.colorbar()  
plt.xticks(range(len(corr.columns)),corr.columns,rotation=90)  
plt.yticks(range(len(corr.columns)),corr.columns)  
plt.show()
```

INTERPRETATION

```
print("Knowledge discovered from the dataset has been interpreted to support hotel business  
decisions.")
```

Output:

	hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	arrival_date_week_number	arrival_date_day_of_month	stays_in_weekend_nights	stays_in_week_nights	adults	...	deposit_type	agent	company	d
0	Resort Hotel	0	342	2015	July	27	1	0	0	2	...	No Deposit	NaN	NaN	
1	Resort Hotel	0	737	2015	July	27	1	0	0	2	...	No Deposit	NaN	NaN	
2	Resort Hotel	0	7	2015	July	27	1	0	1	1	...	No Deposit	NaN	NaN	
3	Resort Hotel	0	13	2015	July	27	1	0	1	1	...	No Deposit	304.0	NaN	
4	Resort Hotel	0	14	2015	July	27	1	0	2	2	...	No Deposit	240.0	NaN	

5 rows × 32 columns

dtypes: float64(4), object(6)
memory usage: 7.9+ KB
None

	Patient_ID	Age	Gender	Blood_Pressure	Cholesterol	BMI
count	100	100.000000	100	100.000000	100.000000	100.000000
unique	100	NaN	5	NaN	NaN	NaN
top	P1000	NaN	Male	NaN	NaN	NaN
freq	1	NaN	37	NaN	NaN	NaN
mean	NaN	47.168675	NaN	137.093023	226.750000	32.228750
std	NaN	12.394866	NaN	12.380033	38.606405	5.571885
min	NaN	25.000000	NaN	120.000000	180.000000	22.500000
25%	NaN	35.000000	NaN	130.000000	200.000000	27.800000
50%	NaN	47.168675	NaN	137.093023	220.000000	32.228750
75%	NaN	55.000000	NaN	150.000000	250.000000	35.400000
max	NaN	65.000000	NaN	160.000000	300.000000	40.100000

	Smoker	Diagnosis	Admission_Date	Notes
count	100	100	100	100
unique	4	2	88	3
top	Yes	Diabetes	2023-10-30	Follow-up required
freq	43	75	2	68
mean	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN

```
<class 'pandas.core.frame.DataFrame'>
```

```
Index: 87396 entries, 0 to 119389
```

```
Data columns (total 33 columns):
```

#	Column	Non-Null Count	Dtype
0	hotel	87396 non-null	object
1	is_canceled	87396 non-null	int64
2	lead_time	87396 non-null	int64
3	arrival_date_year	87396 non-null	int64
4	arrival_date_month	87396 non-null	object
5	arrival_date_week_number	87396 non-null	int64
6	arrival_date_day_of_month	87396 non-null	int64
7	stays_in_weekend_nights	87396 non-null	int64
8	stays_in_week_nights	87396 non-null	int64
9	adults	87396 non-null	int64
10	children	87396 non-null	float64
11	babies	87396 non-null	int64
12	meal	87396 non-null	object
13	country	87396 non-null	object
14	market_segment	87396 non-null	object
15	distribution_channel	87396 non-null	object
16	is_repeated_guest	87396 non-null	int64
17	previous_cancellations	87396 non-null	int64
18	previous_bookings_not_canceled	87396 non-null	int64
19	reserved_room_type	87396 non-null	object
20	assigned_room_type	87396 non-null	object
21	booking_changes	87396 non-null	int64
22	deposit_type	87396 non-null	object
23	agent	87396 non-null	float64
24	days_in_waiting_list	87396 non-null	int64
25	customer_type	87396 non-null	object
26	adr	87396 non-null	float64
27	required_car_parking_spaces	87396 non-null	int64
28	total_of_special_requests	87396 non-null	int64
29	reservation_status	87396 non-null	object
30	reservation_status_date	87396 non-null	datetime64[ns]
31	total_nights	87396 non-null	int64
32	total_guests	87396 non-null	float64

```
dtypes: datetime64[ns](1), float64(4), int64(17), object(11)
```

```
memory usage: 22.7+ MB
```

```
company      82137
agent       12193
country      452
children      4
reserved_room_type    0
assigned_room_type    0
booking_changes    0
deposit_type    0
hotel           0
previous_cancellations    0
days_in_waiting_list    0
customer_type    0
adr           0
required_car_parking_spaces    0
total_of_special_requests    0
reservation_status    0
previous_bookings_not_canceled    0
is_repeated_guest    0
is_canceled    0
distribution_channel    0
market_segment    0
meal          0
babies        0
adults        0
stays_in_week_nights    0
stays_in_weekend_nights    0
arrival_date_day_of_month    0
arrival_date_week_number    0
arrival_date_month    0
arrival_date_year    0
lead_time     0
reservation_status_date    0
dtype: int64
```

	total_nights	total_guests
0	0	2.0
1	0	2.0
2	1	1.0
3	1	1.0
4	2	2.0

Result: